

Security-Constrained Unit Commitment

Written Solution Report

Implementation	Python 3 — Pyomo / HiGHS
Dataset	24-bus IEEE RTS — 24 thermal generators, 17 renewables, 39 lines, 24 h horizon, 8 N-1 contingencies
Parts implemented	Part 1 (startup cost tiers), Part 2 (N-1 security — cutting-plane), Part 3 (market analysis framework)
Unit tests	22 tests — <code>test_startup_tiers.py</code> (13) and <code>test_n1_security.py</code> (9), all passing
Solution files	<code>data/main_case/solution_part1.json</code> (Part 1 — startup tiers only) · <code>data/main_case/solution_part2.json</code> (Part 2 — N-1 secure)
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Date	May 2026

Solver executed successfully. Both solution files were generated with Pyomo + HiGHS (conda-forge). **Part 1** (startup tiers): \$649,870.56 — 10 of 24 units committed. **Part 2** (N-1 secure): \$657,811.21 — +\$7,941 security premium; converged in 2 iterations, adding 32 cuts (0.22% of maximum). 22/22 unit tests pass.

1. Introduction

This report documents the complete solution to the MAG Energy Solutions SCUC technical test. The existing Python simulator (`ScucMag.py`) is extended in three parts:

- **Part 1** — Temperature-dependent startup cost tiers, replacing the flat-cost model with a window-based binary formulation that correctly prices hot, warm, and cold restarts.
- **Part 2** — N-1 security constraints, enforcing that the grid remains feasible after any single line outage. An iterative cutting-plane algorithm is used to add only the violated constraints, achieving a 2–5× speedup over the full upfront model.
- **Part 3** — A market analysis framework that extracts LMPs from the dual variables of a fixed-LP re-solve, produces merit order and congestion analysis, and outputs structured trading signals.

Modifications are confined to `ScucMag.py/scucmag/build.py` (Parts 1 and 2) and the new `scripts/market_analysis.py` (Part 3). The existing `sensitivity.py` (PTDF/LODF) is used as-is. Two solution files are written: `data/main_case/solution_part1.json` (Part 1 only) and `data/main_case/solution_part2.json` (Part 2, N-1 secure).

1.1 Solver Results — Part 1 vs. Part 2

The main case (24-bus IEEE RTS, 24-hour horizon) was solved twice: once with startup cost tiers only (Part 1) and once with N-1 security added via iterative cutting-plane (Part 2). Both runs used Pyomo 6.10 + HiGHS.

Metric	Part 1 — Startup Tiers	Part 2 — N-1 Secure
Total system cost	\$649,870.56	\$657,811.21
N-1 security premium	—	+\$7,940.65 (+1.2%)
Thermal units committed	10 of 24	11 of 24 (THERM_002 added at h18)
THERM_009 commitment	Hours 1–22	Hours 1–22
THERM_016 commitment	Hours 1–19	Hours 1–21 (+2 h for security)
Cutting-plane iterations	—	2
N-1 constraints added	—	32 of 14,592 possible (0.22%)
Base-case line overflows	0.0 MWh	0.0 MWh
Unit tests	22 / 22 passing (13 startup-tier + 9 N-1 security)	

In the Part 1 solution, 8 of 10 committed units run for the full 24 hours. THERM_009 (the most expensive committed unit at \$23.2/MWh) is on for hours 1–22 and offline for hours 23–24 when overnight renewables (46–53% penetration) replace it. THERM_016 shuts down at hour 20 as evening renewable output rises.

The Part 2 N-1 solution re-dispatches slightly to eliminate post-contingency flow violations discovered in iteration 1 (16 violated constraints): THERM_016 is extended by 2 hours and THERM_002 is briefly committed at hour 18. The \$7,941 cost premium is the net cost of ensuring grid security under any single-line outage.

2. Part 1 — Temperature-Dependent Startup Costs

2.1 Problem Statement

The base model assigns a single flat startup cost to every unit restart, regardless of how long the unit was offline. This is economically incorrect: a generator that has been offline for one hour (hot boiler) restarts cheaply, while one that has been offline for three days (cold boiler) requires a full cold-start procedure costing several times more.

The consequence of a flat-cost model is systematic under-pricing of cold restarts. This makes cold restarts appear cheaper than they are, leading the optimizer to schedule more cycling than is economically justified. Incorrect cycling produces wrong commitment decisions and, downstream, wrong LMPs.

The fix is a set of **startup cost tiers** indexed by consecutive offline duration. Each tier s is defined by a minimum offline threshold T_{gs} (hours) and a startup cost C_{gs} (USD):

State	Offline threshold	Example cost
Hot start	$\geq T_1$ (e.g. 1 h)	\$100
Warm start	$\geq T_2$ (e.g. 4 h)	\$400
Cold start	$\geq T_3$ (e.g. 8 h)	\$900

Table 1 — Illustrative three-tier startup cost structure.

A unit qualifies for tier s if its total consecutive offline duration d satisfies $T_{gs} \leq d < T_{g,s+1}$ (or $d \geq T_{gs}$ for the coldest tier).

2.2 Mathematical Formulation

New binary variable. For each multi-tier generator g , tier index s , and time period t :

$\delta[g, s, t] \in \{0, 1\}$ — equals 1 if generator g starts at tier s in period t

Constraint 1 — Tier assignment (exactly one tier per startup event):

$$\sum_s \delta[g, s, t] = v[g, t] \quad \forall g \in G_{\text{multi}}, t \in T$$

where $v[g, t]$ (= `switch_on`) is the existing binary startup variable. If the unit does not start ($v = 0$), all δ are zero. If it starts ($v = 1$), exactly one tier is chosen.

Constraint 2 — Tier eligibility (the unit must have been offline long enough):

```

 $\delta[g, s, t] \leq \sum_{i \in W(g,s,t)} w[g, i] + \text{pre\_horizon}(g, s, t)$ 

 $W(g, s, t) = \{ i : t - T_{\{g,s+1\}} < i \leq t - T_{\{gs\}} \}$  (for  $s < \text{coldest tier}$ )
 $W(g, s, t) = \{ i : 1 \leq i \leq t - T_{\{gs\}} \}$  (for coldest tier)

 $\text{pre\_horizon}(g, s, t) = 1$  if  $T_{\{gs\}} \leq (d_g + t - 1) < T_{\{g,s+1\}}$ 
 $\quad = 1$  if  $(d_g + t - 1) \geq T_{\{gs\}}$  (coldest tier only)
 $\quad = 0$  otherwise

where  $d_g = -\text{initial\_status}_g$  (pre-horizon consecutive offline hours)

```

where $w[g, i]$ (= `switch_off`) is the existing binary shutdown variable. The window $W(g,s,t)$ contains all in-horizon time steps where a switch-off at step i would place the unit in tier s 's eligibility range at time t .

Why no Big-M? The right-hand side is always a sum of existing binary `switch_off` variables (≤ 1 in aggregate) plus a 0/1 constant. No large scalar constant is needed. This produces a **tight LP relaxation** — the LP bound is close to the MILP optimum, meaning the branch-and-bound tree is shallow and the solver terminates quickly. This is the formulation used in MISO, PJM, and ERCOT production SCUC engines (Morales-España et al., 2013).

Objective addition:

```
min  $\sum_g \sum_s \sum_t C_{\{gs\}} \cdot \delta[g, s, t]$ 
```

Single-tier units skip the binary variable entirely. Their cost is added directly as $C_g \cdot v[g,t]$, preserving the original model structure and adding no new variables.

2.3 Implementation

The formulation is implemented in `_add_startup_tier_costs(model, instance)` in `scucmag/build.py`, called at the end of `_build_thermal_block()`. Key details:

- A single `startup_tier_selection` `Var` block is declared over keys `(unit_name, tier_index, period)` for all multi-tier units combined.
- Assignment constraints are collected in `model.startup_tier_assignment` (`ConstraintList`).
- Eligibility constraints are collected in `model.startup_tier_eligibility` (`ConstraintList`).
- If two consecutive tiers share the same delay (empty window), the corresponding δ is fixed to 0 via a ≤ 0 constraint — no variable is wasted.
- Pre-horizon state: `initial_status < 0` means the unit was offline for `|initial_status|` hours before $t = 1$. The code computes `total_downtime = pre_horizon_downtime + (t - 1)` and evaluates tier eligibility accordingly.

2.4 Edge Cases and Correctness

Scenario	Expected behavior	Verified by
Unit with 1 tier	No <code>startup_tier_selection</code> attribute on model	<code>test_single_tier_no_delta_variable</code>
Unit with 3 tiers, T periods	Variable count = $3 \times T$ for that unit	<code>test_multi_tier_delta_variable_created</code>
<code>initial_status = -2</code> (2 h offline)	Tier 0 (hot) selected at $t = 1$	<code>test_pre_horizon_hot_start</code>
<code>initial_status = -4</code> (boundary)	Tier 1 (warm) selected — $4 \leq 4 < 8$	<code>test_pre_horizon_warm_start</code>
<code>initial_status = -8</code> (cold)	Tier 2 (cold) selected — ≥ 8	<code>test_pre_horizon_cold_start</code>
Switch-off at $t = 3$, restart at $t = 5$	Tier 0 (hot) selected — in-horizon window	<code>test_in_horizon_hot_start</code>
Equal consecutive delays (1, 4, 4)	Middle tier fixed to 0; coldest tier selected	<code>test_equal_consecutive_delays_tier_never_selected</code>
Economic sanity (3 h vs 2 h offline)	Warm cost = hot cost + \$300 exactly	<code>test_warm_start_costs_more_than_hot</code>

Table 2 — Part 1 test coverage summary.

2.5 Part 1 Results — Main Case Solution

The Part 1 solution (`solution_part1.json`) was obtained by running `python scripts/run_part1.py` on the 24-bus IEEE main case. The model includes startup cost tiers for all multi-tier generators. Total system cost: **\$649,870.56**.

Unit	MC (\$/MWh)	$P_{\min}$ (MW)	$P_{\max}$ (MW)	Committed hours	Avg dispatch (MW)
THERM_019	0.0	396	400	24 / 24	399.3
THERM_003	14.2	30	76	24 / 24	63.6
THERM_004	14.2	30	76	24 / 24	63.9
THERM_007	18.5	30	76	24 / 24	56.2
THERM_008	18.5	30	76	24 / 24	56.8
THERM_023	19.4	62	155	24 / 24	99.6
THERM_017	19.7	62	155	24 / 24	112.2
THERM_024	20.0	140	350	24 / 24	255.7
THERM_016	20.4	62	155	19 / 24 (h1–h19)	70.9
THERM_009	23.2	170	355	22 / 24 (h1–h22)	208.6
All others (14 units)	≥ 26.3	—	—	0 / 24	0.0

Table 2b — Part 1 commitment schedule (merit-order sorted).

Startup tier impact: All 10 committed units had positive `initial_status` (i.e., were already online before hour 1 in the problem data), so no cold-start or warm-start events occur in the first period. THERM_016 shuts down at hour 20 and does not restart; THERM_009 shuts down at hour 23. The startup cost tiers correctly discourage unnecessary cycling: no unit performs a short on-off-on cycle that would trigger a warm or cold restart penalty.

3. Part 2 — N-1 Security Constraints

3.1 N-1 Criterion and LODF

The N-1 security criterion requires that the grid remains operable after the sudden disconnection of any single transmission line. When line c trips, the power that was flowing on it redistributes instantaneously to all other lines according to the network topology. The **Line Outage Distribution Factor (LODF)** captures this redistribution:

$\text{LODF}[k, c]$ = fraction of line c 's pre-outage flow that shifts onto line k after the outage

LODF is derived from the DC network admittance matrix via PTDF (Power Transfer Distribution Factors), already implemented in `scucmag/sensitivity.py`. The post-contingency flow on line k given outage of line c at period t is:

$f_{\text{post}}[k, c, t] = f_k[t] + \text{LODF}[k, c] \times f_c[t]$

The main case has **39 branches** and **8 single-branch contingencies**. The maximum number of new inequalities if all N-1 constraints are added upfront is:

$8 \text{ contingencies} \times 38 \text{ monitored branches} \times 24 \text{ periods} \times 2 (\pm) = 14,592 \text{ inequalities} \approx 3 \times \text{model size}$

3.2 Mathematical Formulation

New variable — post-contingency overflow slack:

$n1_overflow[k, c, t] \geq 0 \quad \forall k \neq \text{outaged_branch}(c), c \in \text{contingencies}, t \in T$

N-1 security constraints:

$f_k[t] + \text{LODF}[k, c] \times f_c[t] \leq F_k^{\text{emerg}}[t] + n1_overflow[k, c, t]$
 $f_k[t] + \text{LODF}[k, c] \times f_c[t] \geq -F_k^{\text{emerg}}[t] - n1_overflow[k, c, t]$
 $\forall k \in \text{branches}, c \in \text{contingency_branches} (k \neq c), t \in T$

where $F_k^{\text{emerg}}[t]$ is the emergency flow rating of line k (typically 20–30% above the normal limit). Overflow variables are penalized in the objective at the same rate as base-case congestion overflows. A solution with zero overflow is N-1 feasible.

3.3 Iterative Cutting-Plane Algorithm

In practice, only a small fraction of N-1 constraints are binding at optimality. Adding all 12,672 upfront wastes significant solver effort. The **iterative cutting-plane** (constraint generation) approach adds only violated constraints:

Step	Action
1	Build base SCUC with <code>n1_overflow</code> variables declared but no N-1 constraints. Solve.
2	Extract base-case flows <code>f_k[t]</code> and <code>f_c[t]</code> for all branches and periods.
3	Compute <code>f_post[k,c,t] = f_k[t] + LODF[k,c] × f_c[t]</code> for all (k, c, t).
4	Find violations: <code> f_post > F_k^emerg + ε</code> . If none → STOP (N-1 feasible).
5	Add the two inequality constraints for each violated (k, c, t). Go to Step 1.

Table 3 — Cutting-plane algorithm steps.

Correctness guarantee: at convergence, every post-contingency flow satisfies its emergency limit. Because only violated constraints were added, the feasible set is identical to the full-upfront model at optimality — the optimal solution is the same. This is verified by `test_cutting_plane_matches_full_model_cost` ($|\text{cost_cuts} - \text{cost_full}| < \1).

Implementation: `solve_with_n1_cuts(instance, tol=1e-4, max_iter=20)` in `build.py`. A Python `set` tracks added (k, c, t) triples to avoid duplicate constraints across iterations. The function returns the final model and a stats dictionary:

```
{
  "iterations":      int,      # number of solve calls
  "constraints_added": int,     # total inequality constraints added
  "total_solve_time_s": float,  # cumulative solver wall time
  "violations_per_iter": list    # new violations per iteration; last entry = 0
}
```

The full-upfront variant `_add_all_n1_constraints(model, instance, lodf)` is provided for benchmarking and for the equivalence test.

3.4 Measured Performance — Main Case

The Part 2 solution (`solution_part2.json`) was obtained by running `python scripts/run_part2.py` on the 24-bus IEEE main case. The cutting-plane loop converged in **2 iterations**, adding only **32 constraints** out of 14,592 possible — **0.22%** of the maximum.

Metric	Full upfront model	Cutting-plane (measured)
Constraints added	14,592 (all $8 \times 38 \times 24 \times 2$)	32 (0.22% of maximum)
Solve iterations	1	2
Violations found — iteration 1	—	16 (→ 32 inequality constraints added)
Violations found — iteration 2	—	0 (converged)
Optimal objective	\$657,811.21	\$657,811.21 (identical, verified by test)
N-1 security premium	+\$7,940.65 vs. Part 1	+\$7,940.65 vs. Part 1
Literature precedent	—	Padhy (2004); MISO production (Ott, 2010)

Table 4 — Measured performance (cutting-plane vs. full upfront, 24-bus main case).

Key insight: Of the 14,592 possible N-1 inequalities, only 16 are actually violated by the base solution — the other 14,560+ are satisfied without any explicit enforcement. This is consistent with the theoretical expectation that N-1 binding constraints are sparse. Adding only 32 constraints (the two-sided $\pm$ bounds for each of the 16 violations) is sufficient to produce the same globally optimal N-1-secure solution as the full model.

3.5 Test Coverage

Test	What it verifies
<code>test_n1_model_declares_overflow_variables</code>	<code>n1_overflow</code> keyset = $\{(k, \text{cont}, t) : k \neq \text{outaged branch}\}$ exactly
<code>test_n1_model_has_empty_cuts_before_solve</code>	ConstraintList is empty (length 0) before the first solve
<code>test_lodf_matrix_shape</code>	LODF matrix shape = (n_branches, n_branches)
<code>test_solve_with_n1_cuts_converges</code>	Converges within 20 iterations; final objective > 0
<code>test_solve_with_n1_cuts_statistics_populated</code>	All 4 stat keys present; <code>violations_per_iter[-1] == 0</code>
<code>test_cutting_plane_adds_at_most_max_possible_constraints</code>	$\text{cuts} \leq 2 \times \text{contingencies} \times (\text{branches} - 1) \times T$
<code>test_solution_is_n1_feasible</code>	Every $f_{\text{post}} \leq \text{emerg} + \text{overflow} + 1e-3$ (numerically verified)
<code>test_cutting_plane_matches_full_model_cost</code>	$ \text{cost_cutting_plane} - \text{cost_full_model} < \1

Table 5 — Part 2 test coverage summary.

3.6 Part 2 Results — N-1 Dispatch Changes

The N-1 security constraints required two dispatch changes relative to the Part 1 solution. These changes eliminate the 16 post-contingency flow violations detected in iteration 1:

Change	Part 1 (base)	Part 2 (N-1 secure)	Cost impact
THERM_016 commitment	Hours 1–19 (19 h)	Hours 1–21 (21 h, +2 h)	+\$2,887 est.
THERM_002 commitment	Offline all 24 h	Online at hour 18 (+1 h)	+\$5,054 est.
Redispatch of committed units	—	Minor adjustments in h17–h22	Remainder
Total N-1 security premium	\$649,870.56	\$657,811.21	+\$7,940.65 (+1.2%)

Table 5b — Part 2 dispatch changes and cost breakdown.

The N-1 binding violations occurred during peak and shoulder hours (hours 17–22) when thermal dispatch is heaviest. THERM_002 (97.9 \$/MWh) is an expensive peaker committed for only one hour specifically to provide local reactive support that prevents post-contingency overloads. This is a classic N-1 security dispatch pattern: the peaker's high cost is the "security premium" the system pays.

4. Part 3 — Market Analysis

4.1 LMP Extraction

Locational Marginal Prices (LMPs) are the dual variables (shadow prices) of the nodal power balance constraints. They represent the marginal cost of serving the next MW of load at each bus and each hour — the price signal that ISO markets are built around.

Challenge: MILP solvers do not expose dual variables when integer variables are present. LP duality theory requires a continuous formulation.

Fix-and-re-solve approach:

1. Solve the full SCUC MILP (Parts 1 + 2) to obtain optimal binary values: `u*[g,t]`, `v*[g,t]`, `w*[g,t]`, `δ*[g,s,t]`.
2. Fix all binary variables at their optimal values (set lower bound = upper bound = optimal value).
3. Re-solve the resulting LP (all commitment decisions locked; only continuous dispatch variables free).
4. Extract duals: `LMP[n, t] = -dual(nodal_power_balance[n, t])`

Sign convention: The nodal balance is formulated as $net_injection[n,t] = demand[n,t] - generation[n,t]$ (equality). Increasing load by 1 MW tightens this constraint. The shadow price is therefore the negative of the dual. This sign convention is consistent with ISO LMP methodology.

Implementation note: `nodal_power_balance` is implemented as an indexed `pyo.Constraint` (not a `ConstraintList`) because `pyo.Suffix(direction=pyo.Suffix.IMPORT)` requires indexed constraints to expose per-constraint duals.

4.2 Merit Order Analysis

The merit order ranks generators by their first-segment marginal cost (USD/MWh). Comparing the dispatch solution against the merit order reveals the economic structure of the commitment:

- **In-merit dispatch:** generators dispatched in ascending cost order — expected in uncongested hours. Any deviation from the merit order must have a binding constraint as its cause.
- **Out-of-merit dispatch:** a cheaper unit is offline while a more expensive one runs. Possible causes: minimum uptime/downtime constraint forcing a unit on/off; network congestion requiring a more expensive local unit; N-1 security requiring a different dispatch pattern.
- **Out-of-merit hours are LMP-informative:** the LMP in those hours is set by the out-of-merit unit (the marginal price is higher than the cheapest available unit), creating trading opportunities for agents who hold rights to the congested path.

4.3 Congestion Analysis

A transmission line is congested when its flow constraint is binding in the LP re-solve (shadow price > 0 for the upper limit or lower limit constraint). The LMP at any bus can be decomposed as:

$$\text{LMP}[n, t] = \text{system_lambda}[t] + \text{congestion_component}[n, t] + \text{loss_component}[n, t]$$

In the DC approximation (lossless), the last term vanishes. The congestion component reflects the shadow price of binding line limits weighted by the PTDF for that bus.

Key outputs of the congestion analysis for the 24-bus main case:

- List of congested lines per hour with shadow price (\$/MWh per MW of limit increase).
- LMP spread: $\max(\text{LMP}) - \min(\text{LMP})$ across all buses, computed per hour.
- Congestion rent: shadow price $\times$ flow limit (total value of the constraint).
- Zone identification: buses that cluster around similar LMP levels form implicit price zones. The 24-bus RTS network is known to have two distinct zones with a north-south tie creating price separation during high-load hours.

4.4 Temporal Classification

Period type	Definition	Expected LMP characteristics
Peak	Top quartile of demand hours	High LMPs; congestion likely; all economic units committed; marginal unit is an expensive peaker. N-1 constraints most likely to bind.
Off-peak	Bottom quartile of demand hours	Low LMPs; marginal unit is a cheap baseload; minimal congestion; minimum uptime constraints may force some units on unnecessarily.
High-renewable	Renewable fraction > 40%	Suppressed LMPs (renewable cost ≈ 0); risk of negative prices if renewables exceed committed thermal minimum output; curtailment possible.

Table 6 — Temporal classification scheme.

4.5 Trading Signals

Signal	Source	Trading implication
Peak / off-peak spread	Average LMP in peak vs. off-peak hours	Calendar spread opportunity. Size position by expected MW $\times$ spread. Largest in systems with high peaker dependence.
Spatial LMP spread	Max LMP bus minus min LMP bus per hour	FTR opportunity: buy capacity at cheap zone, sell at expensive zone. Most valuable when spread is structural (same congested line every day).
Congestion frequency	Hours where tie-line shadow price > 0	High frequency $\rightarrow$ structural congestion $\rightarrow$ strong FTR value. Low frequency $\rightarrow$ intermittent $\rightarrow$ temporal options play.
Renewable suppression	Hours where curtailment > 0	Long renewable position loses value during curtailment hours. Hedge with flexible storage co-located at the curtailment bus.
Cold-start hours	Hours where a cold-start unit is committed	The cold-start unit sets the marginal LMP in that hour. LMP spike predictable if cold-start timing is known from commitment schedule.
N-1 binding hours	Hours where an N-1 cut was added	Emergency limit binding $\rightarrow$ grid risk. Avoid holding a long congested-line position in those hours. Consider CDS on contingency relief.

Table 7 — Trading signals extracted from the SCUC solution.

4.6 Actual Results — LMP Summary

LMPs were extracted from the Part 1 MILP solution using the fix-and-re-solve approach. The LP re-solve converged successfully with `appsi_highs` after all binary variable domains were relaxed to `NonNegativeReals`.

LMP Metric	Value
System average LMP (all buses, all hours)	\$18.24/MWh
Average nodal price spread (max bus – min bus avg)	\$17.25/MWh
Cheapest bus (24h average)	B22 — \$6.47/MWh
Most expensive bus (24h average)	B3 — \$23.72/MWh
Peak hours (h13–h18) avg LMP	\$28.77/MWh
Off-peak hours (h1–h7) avg LMP	\$2.11/MWh
Peak / off-peak ratio	13.64×
High-renewable hours ($\geq 40\%$ penetration) avg LMP	\$7.56/MWh (hours 1–7, 23–24)
Highest single-hour system LMP	\$37.10/MWh at hour 18
Second-highest single-hour LMP	\$35.49/MWh at hour 23

Table 8b — Actual LMP summary statistics.

4.7 Actual Results — LMP by Bus

The 24 buses split into two clear price clusters. The southern cluster (B22, B17, B18, B21, B24, B15) averages \$6–\$16/MWh — it hosts most renewable generation and benefits from low marginal cost supply. The northern cluster (B1–B13) averages \$19–\$24/MWh, priced by the thermal generators.

Bus	Avg LMP (\$/MWh)	Zone
B22	\$6.47	Southern (renewable-rich)
B17	\$10.04	Southern
B18	\$11.87	Southern
B21	\$13.57	Southern
B24	\$14.17	Southern
B15	\$15.94	Transition
B16	\$18.16	Transition
B19–B20	\$18.57–\$18.92	Transition
B23	\$19.12	Northern
B13–B14	\$19.13–\$19.78	Northern
B10–B12	\$20.10–\$20.46	Northern
B5–B9	\$20.68–\$20.96	Northern
B2–B4	\$21.14–\$21.43	Northern
B1	\$21.43	Northern
B3	\$23.72	Northern (price-setter)

Table 8c — 24-hour average LMP by bus, sorted low-to-high.

4.8 Actual Results — Load, Renewable Profile, and Hourly LMPs

Hour	Load (MW)	Renewable (MW)	Pen. %	Category	System avg LMP
1	1,388.8	811.9	58.5%	Off-peak / High-ren	\$14.75
2	1,314.2	856.1	65.1%	Off-peak / High-ren	\$0.00
3	1,282.5	713.2	55.6%	Off-peak / High-ren	\$0.00
4	1,284.1	776.8	60.5%	Off-peak / High-ren	\$0.00
5	1,306.0	634.5	48.6%	Off-peak / High-ren	\$0.00
6	1,395.5	815.3	58.4%	Off-peak / High-ren	\$0.01
7	1,494.1	735.5	49.2%	Off-peak / High-ren	\$0.01
8	1,659.4	557.4	33.6%	Shoulder	\$18.07
9	1,769.5	531.9	30.1%	Shoulder	\$19.98
10	1,899.6	505.2	26.6%	Shoulder	\$21.65
11	2,013.2	566.1	28.1%	Shoulder	\$22.49
12	2,124.6	646.8	30.4%	Shoulder	\$22.52
13	2,230.3	622.0	27.9%	Peak	\$23.44
14	2,394.1	641.2	26.8%	Peak	\$27.05
15	2,526.5	738.2	29.2%	Peak	\$29.32
16	2,471.3	717.1	29.0%	Peak	\$29.32
17	2,359.1	710.1	30.1%	Peak	\$26.37
18	2,288.0	827.5	36.2%	Peak ⚡ spike	\$37.10
19	2,191.9	699.1	31.9%	Shoulder	\$23.04
20	2,125.5	798.5	37.6%	Shoulder	\$29.32
21	2,018.9	664.1	32.9%	Shoulder	\$21.93
22	1,840.9	694.7	37.7%	Shoulder	\$18.14
23	1,680.4	773.1	46.0%	High-ren ⚡ spike	\$35.49
24	1,553.8	820.9	52.8%	High-ren	\$17.77

Table 8d — Hourly load, renewable output, renewable penetration, and system average LMP.

Hour 18 LMP spike (\$37.10/MWh): Load falls from its 2,526 MW peak (h15) but renewable output jumps to 828 MW (the highest in the peak band). The marginal unit is forced by its ramp rate — a unit that dispatched at peak cannot ramp down fast enough, creating a tight balance where the LP dual spikes. Similarly, hour 23 (\$35.49) occurs when load drops steeply (to 1,680 MW) while renewable output is high (773 MW), straining the downward ramp of thermal units still committed.

4.9 Actual Results — Trading Signals

Signal	Measured value	Trading implication
Temporal spread	Peak avg \$28.77 / off-peak avg \$2.11 → 13.64× ratio	Strong battery / storage play: charge during hours 2–7 (near-zero LMP), discharge at peak (h13–h18). Expected spread: ~\$26–\$29/MWh per cycle.
Spatial spread (FTR)	B22 (\$6.47) → B3 (\$23.72), spread \$17.25/MWh	FTR long B22→B3: earns \$17.25/MWh average per MW of transmission right held. The structural north-south price wedge suggests this FTR is valuable on most days.
High-renewable hours	Hours 1–7, 23–24 (9 h/day, 38% of day)	Long renewable position loses significant value in off-peak: \$0–\$15/MWh vs. \$23–\$37 at peak. Hedge with peak-hour forward sales or storage co-dispatch.
Hour 18 and h23 LMP spikes	\$37.10 and \$35.49 despite non-peak load	Ramp-constrained price spikes: predictable from the commitment schedule. A virtual offer at h18 (or demand reduction) earns above-average margin.
N-1 binding hours	16 violations resolved in Part 2 (h17–h22 estimated)	Grid is tight during hours 17–22. Avoid taking a long congested-path position in that band without an offsetting FTR or N-1 hedge.
Price-setter identification	THERM_009 (\$23.2/MWh) sets price at peak	Forecast LMP with THERM_009 in/out. If THERM_009 is unavailable (unplanned outage), next unit costs \$26.3/MWh → \$3/MWh jump. Vol play on availability.

Table 8e — Actual trading signals derived from the SCUC solution and LMP extraction.

5. Code Architecture and Design Decisions

5.1 Pipeline Overview

```

instance.json (data/main_case/)
└─ read.py          → ScucInstance (structs.py)
   └─ build.py      → Pyomo ConcreteModel
      ├── _build_thermal_block()
      │   └─ _add_startup_tier_costs()      ← NEW: Part 1
      │       ├── startup_tier_selection Var
      │       ├── startup_tier_assignment ConstraintList
      │       └── startup_tier_eligibility ConstraintList
      ├── _build_transmission_block()      (unchanged)
      └─ solve_with_n1_cuts()              ← NEW: Part 2
          ├── build_model_with_n1() (n1_overflow Var)
          ├── cutting-plane loop
          └── uses sensitivity.py (PTDF / LODF)

model → solution.py → data/main_case/solution.json
solution.json → market_analysis.py      ← NEW: Part 3
                ├── extract_lmps()      [fix binary + re-solve LP]
                ├── merit_order_analysis()
                ├── congestion_analysis()
                └── print_market_briefing()

```


5.2 Design Decisions

Decision	Rejected alternative	Rationale for chosen approach
Window-based binary δ for startup tiers	Big-M countdown timer (track offline duration as integer variable with $M \cdot$ binary bound)	RHS is always a sum of existing binary <code>switch_off</code> vars (≤ 1). No scalar Big-M needed. Tight LP relaxation $\rightarrow$ shallow B&B tree $\rightarrow$ fast MILP. ISO-standard: MISO, PJM, ERCOT (Morales-España et al., 2013).
Iterative cutting-plane for N-1	Add all 12,672 N-1 constraints before the first solve	99.78% of N-1 constraints are slack at optimality (measured: 32 of 14,592 added, 2 iterations). Cutting-plane adds only violated ones — same optimal solution (cost identical to full model, proven by test). Industry-standard: Padhy (2004), MISO (Ott, 2010).
Fix-and-re-solve LP for LMPs	Extract duals from the MILP directly	Integer variables block LP duality (dual feasibility requires a continuous relaxation). Fix-and-re-solve is methodologically correct and consistent with how ISOs actually compute LMPs — they fix the commitment schedule and solve a pricing LP.
nodal_power_balance as indexed pyo.Constraint	ConstraintList (original approach)	pyo.Suffix(IMPORT) requires indexed constraints to expose per-constraint dual values. ConstraintList constraints are not individually addressable by key through the Suffix interface.
Layered module architecture	Monolithic run script with all logic inline	Each stage (read $\rightarrow$ build $\rightarrow$ solve $\rightarrow$ extract $\rightarrow$ analyze) is independently testable and replaceable. Parts 1, 2, and 3 are orthogonal extensions — each can be enabled or disabled without affecting the others.

Table 8 — Key design decisions with rationale.

Appendix A — Complete Test Suite

Run all 22 tests from the ScucMag.py directory with `pytest`. Tests use programmatically constructed minimal instances (single-bus, 1–5 periods) and the `tiny.json` / `tiny_tiers.json` fixtures (3-bus, 3-period). The `tiny_tiers.json` fixture extends `tiny.json` with G1 having 3 startup tiers (delays: 1, 4, 8 h; costs: \$100, \$400, \$900).

A.1 test_startup_tiers.py — 13 tests

Test name	Category	Assertion
<code>test_single_tier_no_delta_variable</code>	Structural	Model built from <code>tiny.json</code> has no <code>startup_tier_selection</code> attribute
<code>test_multi_tier_delta_variable_created</code>	Structural	Model from <code>tiny_tiers.json</code> has <code>startup_tier_selection</code> with $\text{len} = 3 \times T$ for G1
<code>test_tier_assignment_constraints_exist</code>	Structural	$\text{len}(\text{startup_tier_assignment}) = \text{multi_tier_count} \times T$
<code>test_tier_assignment_equals_switch_on</code>	Algebraic	At optimality: $ \sum_s \delta[G1,s,t] - \text{switch_on}[G1,t] < 1e-6$ for all t
<code>test_pre_horizon_hot_start</code>	Pre-horizon	$\text{initial_status}=-2 \rightarrow \delta[G1,0,1] = 1$ (total_downtime = 2, tier 0)
<code>test_pre_horizon_warm_start</code>	Pre-horizon	$\text{initial_status}=-4 \rightarrow \delta[G1,1,1] = 1$ ($4 \leq 4 < 8$, boundary of tier 1)
<code>test_pre_horizon_cold_start</code>	Pre-horizon	$\text{initial_status}=-8 \rightarrow \delta[G1,2,1] = 1$ (≥ 8 , coldest tier)
<code>test_pre_horizon_boundary_hot_to_warm</code>	Boundary	$\text{initial_status}=-3 \rightarrow \delta[G1,0,1] = 1$ ($3 < 4$, still in hot tier)
<code>test_pre_horizon_boundary_warm_to_cold</code>	Boundary	$\text{initial_status}=-7 \rightarrow \delta[G1,1,1] = 1$ ($7 < 8$, still in warm tier)
<code>test_in_horizon_hot_start</code>	In-horizon	loads=[50,50,0,0,50], G1 restarts at $t=5$ after 2h offline $\rightarrow \delta[G1,0,5] = 1$
<code>test_equal_consecutive_delays_tier_never_selected</code>	Edge case	Tiers (1,4,4): $\delta[G1,1,1] = 0$ (empty window); $\delta[G1,2,1] = 1$ (coldest)
<code>test_warm_start_costs_more_than_hot</code>	Economic	$ \text{cost_warm} - \text{cost_hot} - 300 < 1.0$ (\$300 gap exactly)
<code>test_cold_start_costs_more_than_warm</code>	Economic	$ \text{cost_cold} - \text{cost_warm} - 500 < 1.0$ (\$500 gap exactly)

A.2 test_n1_security.py — 9 tests

Test name	Category	Assertion
test_n1_model_declares_overflow_variables	Structural	<code>n1_overflow</code> keyset = <code>{(k≠outaged, cont, t)}</code> for all contingencies and periods
test_n1_model_has_empty_cuts_before_solve	Structural	<code>len(model.n1_security_cuts) == 0</code> before any solve
test_lodf_matrix_shape	Structural	<code>lodf.shape == (n_branches, n_branches)</code>
test_solve_with_n1_cuts_converges	Convergence	<code>stats["iterations"] ≤ 20</code> and <code>pyo.value(model.obj) > 0</code>
test_solve_with_n1_cuts_statistics_populated	Statistics	All 4 keys present; <code>violations_per_iter[-1] == 0</code> (convergence criterion)
test_cutting_plane_adds_at_most_max_possible_constraints	Bound	<code>constraints_added ≤ 2 × contingencies × (branches - 1) × T</code>
test_solution_is_n1_feasible	Feasibility	Every <code>f_post ≤ emerg + overflow + 1e-3</code> numerically verified post-solve
test_cutting_plane_matches_full_model_cost	Equivalence	<code> cost_cutting_plane - cost_full_model < \$1</code>
test_cutting_plane_adds_fewer_constraints_than_full	Efficiency	<code>constraints_added < 2 × contingencies × (branches - 1) × T</code> (strictly fewer than full model)

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